MAgent Theory 2.0

Gauge-Covariant Agents
and Inherited Organization

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Abstract

An agent is represented by a local probability law together with mechanisms for receiving records, making predictions, and changing that law. Gauge transformations change its presentation. Interactions communicate selected information between agents and may depend on their current belief and model laws. We formulate these objects on associated statistical bundles, give explicit law-dependent links, and distinguish a normalized comparison likelihood from a direct divergence between peer laws. A coupled two-sector variational dynamics follows from a declared potential and the Fisher metric. Exact conditional-information identities describe what communication preserves and what a receiver supplies through its own model. A parent is then characterized by the information required to inherit the constituents' future evolution and boundary responses. Exact quotient criteria, a finite recursive example, and counterexamples identify this requirement without approximation estimates. The construction of persistent blocks selected by the same dynamics, followed by further inherited formation, remains the central open problem.

Reading the construction

The mathematical objects are introduced in the order in which they are needed: agent laws and records, interactions, dynamics, predictive compression, and inherited organization. The main results include their decisive proofs. The appendices give the internal history construction, the relation between directional links and a base connection, and a notation table. Secondary calculations are collected in a separate supplement.

Definitions specify the model's objects; constitutive choices specify how those objects interact. A theorem states a consequence of explicit hypotheses. An open problem identifies a further construction that is not supplied by those hypotheses. General probability measures are used whenever measurability suffices. Finite alphabets, smooth statistical families, and vector fibers enter only where an example or a differential calculation requires them. No Gaussian closure is assumed.

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Chapter 1

Agents, Possibilities, and Records

The purpose of the theory is to describe agents whose interactions can produce further agents. The first distinction is between an agent's probability law and the possibilities represented by that law. The second is between comparing two descriptions and specifying an interaction that changes them. A parent construction must preserve enough of both to continue the dynamics.

1.1 Statistical sections

Let $\mathcal{C}$ be a smooth base of contexts and let $P \to \mathcal{C}$ be a principal G-bundle. A context labels where a description applies; it is not identified with physical position or time. Let K and M be standard Borel spaces, with measurable G-actions by bimeasurable bijections. Their elements k and m represent state contents and model presentations. The actions may be different representations of the same group. They act on probability measures by pushforward.

Definition 1.1 (Statistical agent section). Set $\mathsf{Y} = \mathsf{K} \times \mathsf{M}$, with the product action, and give $\mathcal{P}(\mathsf{Y})$ the sigma-algebra generated by evaluation on measurable sets. The associated statistical bundle is

$$\mathcal{B} = P \times_G \mathcal{P}(\mathsf{Y}), \qquad (ug, \zeta) \sim (u, g_{\#}\zeta). \tag{1.1}$$

The statistical component of agent i is a section ζ_i over a domain $\mathcal{U}_i \subseteq \mathcal{C}$. Its full state also includes the internal variables and access mechanisms read by its update rule.

At a fixed context, disintegration gives

$$\zeta_i(dk, dm) = q_i(dk \mid m) s_i(dm), \qquad q_i(dk) = \int q_i(dk \mid m) s_i(dm). \tag{1.2}$$

Here q_i is the state marginal and s_i is the law over model presentations. Regular conditional laws exist on the stated spaces, with versions defined up to the appropriate marginal null sets [2]. A model presentation acquires predictive content through a normalized kernel; its name or coordinate value alone supplies no prediction.

The product restriction $\zeta_i = q_i \otimes s_i$ removes dependence between the two sectors within an agent. A product over agents makes a further restriction on represented contents. Neither restriction prevents the update of one law from depending on other laws. Coupling is a property of the dynamics; factorization is a property of a chosen probability description.

Smoothness is an additional hypothesis. When derivatives of laws are required, use a declared regular statistical family stable under the relevant coordinate changes. The whole space $\mathcal{P}(\mathsf{Y})$ is not assigned a smooth manifold structure by notation. Likewise, a collection of pointwise marginals does not determine a probability law on fields over $\mathcal{C}$.

The population at a context consists of the agents whose domains contain that context. Most of the dynamics below is formulated at a common c_*. This retains the ambient bundle and local sections without requiring parallel transport between distinct base points. Independent interactions between agents at c_* are defined in chapter 2. Directional transport within an agent is treated in appendix B.

1.2 Presentation and probability

In local vector coordinates, write $U_i^q : V_q \to E_i^q$ and $U_i^s : V_s \to E_i^s$ for the sector frames. The spaces E_i^q, E_i^s are the represented state and model fibers. A common principal frame can induce both maps through different representations. Reference laws are

$$\bar{q}_i = ((U_i^q)^{-1})_{\#} q_i, \qquad \bar{s}_i = ((U_i^s)^{-1})_{\#} s_i. \tag{1.3}$$

A passive change sends q_i to $(g_i^q)_{\#} q_i$ and U_i^q to $g_i^q U_i^q$, with the corresponding model-sector rule. The reference laws are unchanged. This is a change of description, not an inference step.

For probability measures μ, ν on one measurable space, relative entropy is

$$D_{\mathrm{KL}}(\mu\|\nu) = \int \log \frac{d\mu}{d\nu} d\mu, \tag{1.4}$$

when $\mu \ll \nu$, and is $+\infty$ otherwise. In this formula and below, a displayed integral is understood with its declared measure; no common Lebesgue density is required. Relative entropy compares distinguishability of laws [4].

Proposition 1.2 (Coordinate invariance). For a bimeasurable bijection f, including divergences with value $+\infty$,

$$D_{\mathrm{KL}}(f_{\#}\mu\|f_{\#}\nu) = D_{\mathrm{KL}}(\mu\|\nu). \tag{1.5}$$

Proof. Absolute continuity is preserved in both directions. If $h = d\mu/d\nu$, the transformed density is $h \circ f^{-1}$. Integrating its logarithm against $f_{\#}\mu$ gives the original integral. If absolute continuity fails, a set of zero ν-measure and positive μ-measure is carried to a set with the same property. □

Definition 1.3 (Normalized channel). A Markov kernel $K : X \rightsquigarrow Y$ assigns a probability measure $K(x, \cdot)$ to each x, measurably in x. Its action on a law is

$$(\mu K)(A) = \int K(x, A)\mu(dx). \tag{1.6}$$

A deterministic measurable map $f : X \to Y$ is represented by $K_f(x, \cdot) = \delta_{f(x)}$. Then $\mu K_f = f_{\#}\mu$.

An invertible frame change preserves all distinctions. A selective or noisy channel need not. Both are well-defined operations on general probability laws, but their roles in an agent interaction differ.

1.3 Internal records

At a fixed context, let S_i be the admitted internal-state space of agent i, and let S be the complete configuration space. It contains the statistical laws, source rows, interfaces, and memory variables

that the dynamics reads. Write $\mathrm{pr}_i : \mathsf{S} \to \mathsf{S}_i$ for the local state projection. For the measurable constructions, take these spaces to be standard Borel. A finite collection of probability-law spaces and finite or Euclidean memory spaces supplies such a realization.

Definition 1.4 (Record and readout). A record readout is a measurable map

$$\rho_i : \mathsf{S}_i \longrightarrow \mathsf{O}_i, \tag{1.7}$$

where O_i is the alphabet of internally distinguishable outcomes. If an interaction with source j produces configuration Z_{t+1}, its record variable is

$$O_{ij,t} = \rho_i(\mathrm{pr}_i\, Z_{t+1}). \tag{1.8}$$

For stochastic dynamics this is a measurable function from the probability space of realizations to O_i. A realized value is denoted $o_{ij,t}$. The source label is part of the record only when the internal state retains it.

A record can be transient. Persistence means that the distinction remains accessible through a specified later readout. A list kept outside the agent does not supply this accessibility. A memory coordinate is therefore an explicit part of the model unless the required distinction is already represented in its state.

Proposition 1.5 (Induced record law). Let $\Gamma_{ij} : \mathsf{S} \rightsquigarrow \mathsf{S}$ be a normalized interaction transition. Then

$$\mathcal{R}_{ij}(x, A) = \Gamma_{ij}\left(x, \{x' : \rho_i(\mathrm{pr}_i\, x') \in A\}\right), \tag{1.9}$$

is a normalized record kernel. If $\Gamma_{ij}(x, \cdot) = \delta_{F_{ij}(x)}$, the record is the deterministic function $\rho_i \circ \mathrm{pr}_i \circ F_{ij}$.

Proof. The right side is the pushforward of a probability measure through the measurable map $\rho_i \circ \mathrm{pr}_i$. Its total mass is one, and measurability in x follows from the defining property of a kernel. The deterministic case follows by evaluating a Dirac measure. □

This definition introduces no observing agent outside the population. It specifies where an outcome is available and how it can enter later dynamics. It does not derive the transition Γ_{ij} or decide whether its stochasticity represents objective chance. Those are further modeling choices.

For example, a vector comparison can be represented by

$$r_{ij} : E_i^q \times E_j^q \times \mathbb{R}^r \to \mathbb{R}^r, \qquad r_{ij}(k_i, k_j, \epsilon) = A_i k_i - B_{ij} k_j + \epsilon. \tag{1.10}$$

A noise law turns this map into a record kernel on represented contents. A joint content law determines its predicted outcomes. A transition on agent states must still specify how an outcome is registered and how the agent changes. A predictive likelihood and an internal update thus have different domains.

1.4 What correlations describe

Suppose two represented bits have the independent uniform law. Register the event that they agree. Their conditional joint law assigns probability $1/2$ to each of $(0, 0)$ and $(1, 1)$. Each marginal

remains uniform, while the conditional expectation of their product changes from $1/4$ to $1/2$. An exact joint posterior has acquired dependence that its two marginals cannot report.

This example identifies a concrete need for a joint law: predicting an observable that reads both bits after the shared record. It does not imply that every admissible agent dynamics must propagate that posterior. A coupled product-family dynamics is another precise model, provided it is declared. Similarly, two routes that inspect the same source must share its identity; replacing it by independent redraws specifies a different experiment.

The configuration x of law-valued agents, a sample tuple $(k_i, m_i)_i$ used in a probabilistic model, and an ensemble of possible configurations are three different objects. A deterministic dynamics on x needs no probability law over configurations. A preparation ensemble can be introduced to assess predictions or compression, but it is then stated explicitly. These distinctions will matter when a parent retains a law over contents while discarding information needed to evolve that law.

Chapter 2

Law-Dependent Interactions

An interaction compares or transmits information between agents. Its law may depend on the agents' current statistical states. Gauge covariance specifies how this dependence is represented when coordinates change; it does not require the interaction to be a frame identification or select a unique mechanism. We first construct these interactions at one context. Transport between different contexts is a further construction, developed in appendix B.

2.1 Two sectors and their comparison spaces

Fix a context c and a finite set of agents defined there. Write

$$\nu_i^q = q_i, \qquad \nu_i^s = s_i, \tag{2.1}$$

for the belief and model marginals of agent i. The sector index a takes values in $\{q, s\}$. These laws live on standard Borel spaces E_i^a. No product assumption between the two sectors, or between different agents, is required to define an interaction between their marginals. Such a product is a separate choice when a joint recognition law is needed.

In a vector realization choose reference spaces V_q, V_s and invertible presentation maps

$$U_i^a : V_a \longrightarrow E_i^a, \qquad \bar{q}_i = (U_i^q)_\#^{-1} q_i, \qquad \bar{s}_i = (U_i^s)_\#^{-1} s_i. \tag{2.2}$$

We also write $\bar{\nu}_i^a = (U_i^a)_\#^{-1} \nu_i^a$. The two presentations can come from different representations of the same principal frame. This does not identify the belief and model interaction maps. Their domains, parameters, and dependence on the laws remain separately specified.

A passive re-expression $y_i' = g_i^a y_i$ sends ν_i^a to $(g_i^a)_\# \nu_i^a$ and U_i^a to $g_i^a U_i^a$. Therefore the reference laws in eq. (2.2) are unchanged. A common reference change $U_i^a \mapsto U_i^a h_a$ instead sends every reference law to $(h_a^{-1})_\# \bar{\nu}_i^a$. The first operation changes an agent's displayed coordinates; the second changes the common coordinates in which interaction parameters are written.

Definition 2.1 (Normalized interaction channel). A sector channel from source j to receiver i is a Markov kernel

$$\mathcal{K}_{ij}^a : E_j^a \rightsquigarrow E_i^a. \tag{2.3}$$

It sends a law to $(\nu_j^a \mathcal{K}_{ij}^a)(A) = \int \mathcal{K}_{ij}^a(y, A) \nu_j^a(dy)$. When the channel depends on the agent configuration, this formula is evaluated at that configuration. Covariance requires

$$\mathcal{K}_{ij}^{a\prime}(g_j^a y, A) = \mathcal{K}_{ij}^a(y, (g_i^a)^{-1} A). \tag{2.4}$$

Composition of sending, transmission, and receiving kernels produces another normalized channel by iterated integration. Invertibility is not required. Alternatively, two channels can send the receiver and source laws into a common message space, where their communicated predictions are compared.

Proposition 2.2 (Covariant law comparison). Under eq. (2.4), the divergence

$$D_{ij}^{a} = D_{\mathrm{KL}}(\nu_i^a \| \nu_j^a \mathcal{K}_{ij}^a), \tag{2.5}$$

is unchanged by passive bimeasurable coordinate changes, including when its value is infinite.

Proof. Changing the integration variable in the kernel gives $\nu_j^{a\prime}\mathcal{K}_{ij}^{a\prime} = (g_i^a)_{\#}(\nu_j^a \mathcal{K}_{ij}^a)$. For any bimeasurable bijection g, absolute continuity is preserved and the transformed Radon–Nikodym derivative is $(d\mu/d\eta) \circ g^{-1}$. Its logarithmic integral is therefore unchanged. Failure of absolute continuity is also preserved, giving infinity on both sides. □

This is the coordinate invariance of relative entropy [4]. It makes the comparison intrinsic without asserting that either law is a posterior under a fixed population model.

2.2 Sending, receiving, and exponential links

Choose a message space H_a and reference interfaces

$$T_j^a : V_a \longrightarrow H_a, \qquad O_i^a : H_a \longrightarrow V_a. \tag{2.6}$$

Their direct content map is $M_{ij}^a = U_i^a O_i^a T_j^a (U_j^a)^{-1}$. A deterministic channel can push the entire law through this map even when the image is lower dimensional. A density formula for a nonsingular law cannot then be used on the original ambient space without accounting for that change of support.

Definition 2.3 (Law-dependent reference generator). Let $\mathcal{L}_{ij}^a$ be a measurable function of the reference laws, and of any explicitly declared interface data, with values in $\mathrm{End}(V_a)$. Define

$$L_{ij}^a = \mathcal{L}_{ij}^a(\bar{\nu}_i^a, \bar{\nu}_j^a), \qquad \Theta_{ij}^a = U_i^a \exp(L_{ij}^a)(U_j^a)^{-1}. \tag{2.7}$$

The primitive self-comparison is $L_{ii}^a = 0$. For other edges a receiving–sending factorization is $L_{ij}^a = O_i^a(\bar{\nu}_i^a) T_j^a(\bar{\nu}_j^a)$, provided both maps refer to the same H_a and V_a.

The exponential is taken on the reference endomorphism, not on a map with different domain and codomain. These are $\mathrm{GL}(V_a)$ interaction maps. A smaller group acting on the frames can still re-express them covariantly. Interpreting them as parallel transports of a connection for that smaller group would additionally require generators in its represented Lie algebra.

Proposition 2.4 (Endpoint covariance and reference consistency). The links in eq. (2.7) obey

$$\Theta_{ij}^{a\prime} = g_i^a \Theta_{ij}^a (g_j^a)^{-1}. \tag{2.8}$$

They are also independent of a common reference rechoice if

$$\mathcal{L}^{a\prime}_{ij}((h_a^{-1})_{\#}\bar{\nu}^a_i, (h_a^{-1})_{\#}\bar{\nu}^a_j) = h_a^{-1} L^a_{ij} h_a, \tag{2.9}$$

where the prime includes the re-expression of any supplied interface data.

Proof. Passive site changes leave the reference laws and generator unchanged, so multiplication by the transformed endpoint frames proves eq. (2.8). Under a reference rechoice, the exponential series gives $e^{h_a^{-1} L h_a} = h_a^{-1} e^L h_a$. Substitution cancels both occurrences of h_a in eq. (2.7). □

For factored generators, $O^a_i \mapsto h_a^{-1} O^a_i$ and $T^a_j \mapsto T^a_j h_a$ suffice. Message-coordinate changes $T^a_j \mapsto r_a T^a_j$, $O^a_i \mapsto O^a_i r_a^{-1}$ cancel within the product.

Proposition 2.5 (Selective generator and invertible transport). For finite matrices $O : H \to V$ and $T : V \to H$, the map OT has rank at most $\dim H$, whereas e^{OT} is invertible. Moreover,

$$e^{OT} v = v \quad (v \in \ker T), \qquad \operatorname{im}(e^{OT} - I) \subseteq \operatorname{im} O. \tag{2.10}$$

Proof. The rank statement follows from factorization. The exponential series and $e^{OT} e^{-OT} = I$ prove invertibility. Every positive power of OT vanishes on $\ker T$, and every such power has image in $\operatorname{im} O$. Summing the convergent series proves the remaining assertions. □

Thus the exponential link retains an identity action in unread directions. It is a different operation from sending only the selective content OTv. The identity self-link also differs from a coarse internal loop: a parent may retain nonidentity loop information inherited from several child interactions.

2.3 An explicit constitutive family for general laws

The preceding definition does not select the functional $\mathcal{L}$. The following choice shows that non-constant, covariant generators can depend directly on laws without restricting them to a Gaussian family.

Definition 2.6 (Moment-dependent receiving and sending). For each sector let $V_a = \mathbb{R}^{d_a}$ and admit laws with finite second moments and positive-definite covariance. In reference coordinates write their means and covariances as μ^a_i and M^a_i. For a supplied scalar λ^a_{ij} on each nonself directed edge, put

$$L^a_{ij} = \lambda^a_{ij} \mu^a_i (\mu^a_j)^T (M^a_j)^{-1}, \qquad O^a_{ij}(t) = \lambda^a_{ij} \mu^a_i t, \qquad T^a_j(v) = (\mu^a_j)^T (M^a_j)^{-1} v. \tag{2.11}$$

Here $H_a = \mathbb{R}$, and $L^a_{ij} = O^a_{ij} T^a_j$ has rank at most one. Set $L^a_{ii} = 0$ separately. The edge coefficient is constitutive data, not a consequence of covariance.

Proposition 2.7 (Equivariance of the moment family). The family eq. (2.11) satisfies eq. (2.9) under every invertible linear reference change.

Proof. Under $U^a_i \mapsto U^a_i h_a$, the moments become $\mu^{a\prime}_i = h_a^{-1} \mu^a_i$ and $M^{a\prime}_j = h_a^{-1} M^a_j h_a^{-T}$. Consequently

$$(\mu^{a\prime}_j)^T (M^{a\prime}_j)^{-1} = (\mu^a_j)^T (M^a_j)^{-1} h_a. \tag{2.12}$$

Multiplication with the transformed receiving column gives $L^{a\prime}_{ij} = h_a^{-1} L^a_{ij} h_a$. □

No fixed Euclidean metric was inserted into this comparison: the source covariance supplies the inverse bilinear form and transforms with the law. Finite moments and invertible covariance are assumptions of this example alone. Centered laws give zero generators in this family, so it does not encode all distinctions between laws.

For a non-Gaussian example in two dimensions, take independent centered Laplace coordinates of variance one and translate their joint law to means e_1 and e_2 for two agents. Both covariances equal I. With identity frames and unit edge coefficients,

$$L_{12} = E_{12}, \qquad L_{21} = E_{21}, \qquad \Theta_{12}\Theta_{21} = (I + E_{12})(I + E_{21}) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \tag{2.13}$$

The loop is nontrivial even though both interfaces have rank one. This is a communication loop at one context, not yet the curvature of a base connection.

2.4 Dependence, direction, and inherited responses

Covariance constrains which law-only generators are possible. If a reference transformation fixes both input laws and all supplied data, eq. (2.9) requires its matrix to commute with the resulting generator. Identical centered laws invariant under every orthogonal transformation therefore cannot select a distinguished off-diagonal direction through a fully orthogonally equivariant law-only rule. Directional interfaces can carry additional information, but that information must be declared.

Opposite attending edges are independent constitutive choices. In particular, Θ_{ji} need not equal Θ_{ij}^{-1}, as eq. (2.13) demonstrates. Geometric reversal of the *same* transport is instead its inverse. Endpoint covariance and conjugation of loop products are the familiar transformation laws of lattice link variables [8]; neither imposes an inverse relation between two independently modeled interactions.

When links depend on the current laws, differentiation must include that dependence. At fixed frames a variation of an objective has the form

$$d\mathcal{E} = d_\nu \mathcal{E} + \sum_{a,i,j} \langle d_{L^a_{ij}} \mathcal{E}, D\mathcal{L}^a_{ij}[d\bar{\nu}^a_i, d\bar{\nu}^a_j] \rangle. \tag{2.14}$$

This statement uses a differentiable statistical family and differentiable constitutive maps. It includes both the sender and receiver occurrences. Freezing the link during an update defines another declared dynamics; it is not generally the derivative of the substituted objective. Nor does substituting live recognition laws into a likelihood make that likelihood part of a fixed generative model during recognition variation.

An inherited parent's state must determine its future boundary response, including the law dependence of its generators. If two child configurations with identical retained data give different future responses, those data do not support autonomous closure.

Chapter 3

Coupled inference on agent laws

An interaction between probability laws can be specified directly, without first interpreting it as an observation likelihood. This chapter chooses such an interaction for both belief and model laws, derives its source rows, and supplies a covariant dissipative dynamics. A normalized model of records defines a second construction. Their relation depends on what is held fixed and on whether the interaction compares laws or sampled contents.

3.1 A potential for the two interacting sectors

Fix a context and a finite set of agents. Agent i has state and model spaces $\mathsf{K}_i, \mathsf{M}_i$, both standard Borel. In this chapter its joint law is restricted explicitly to $\zeta_i = q_i \otimes s_i$. Choose a proper model reference r_i and a normalized evaluator $G_i : \mathsf{M}_i \rightsquigarrow \mathsf{K}_i$. The reference joint is

$$R_i(dk, dm) = r_i(dm)G_i(m, dk), \qquad \mathcal{E}_0 = \sum_i D_{\mathrm{KL}}(q_i \otimes s_i \| R_i). \tag{3.1}$$

The evaluator couples the sectors even though their recognition law is a product. Where the terms are finite, disintegration gives

$$D_{\mathrm{KL}}(q_i \otimes s_i \| R_i) = D_{\mathrm{KL}}(s_i \| r_i) + \int D_{\mathrm{KL}}(q_i \| G_i(m, \cdot))s_i(dm). \tag{3.2}$$

This is a conditional relative-entropy identity: the model marginal contributes the first term and the state conditional contributes the second. In particular, the expectation of a log evaluator in this objective is different from the log of the predictive mixture $s_i G_i$.

Let $\mathbf{q} = (q_i)_i$ and $\mathbf{s} = (s_i)_i$. For each allowed directed edge $j \to i$, choose normalized channels $\mathcal{K}^q_{ij}[\mathbf{q}] : \mathsf{K}_j \rightsquigarrow \mathsf{K}_i$ and $\mathcal{K}^s_{ij}[\mathbf{s}] : \mathsf{M}_j \rightsquigarrow \mathsf{M}_i$. Their dependence on the laws is part of the specified interaction. Define

$$\begin{aligned} \nu^q_{ij} &= q_j \mathcal{K}^q_{ij}[\mathbf{q}], \quad D^q_{ij} = D_{\mathrm{KL}}(q_i \| \nu^q_{ij}), \\ \nu^s_{ij} &= s_j \mathcal{K}^s_{ij}[\mathbf{s}], \quad D^s_{ij} = D_{\mathrm{KL}}(s_i \| \nu^s_{ij}). \end{aligned} \tag{3.3}$$

The source sets $\mathcal{N}^q_i, \mathcal{N}^s_i$ are finite and nonempty, with positive normalized reference rows π^q_i, π^s_i. Belief routing uses β_i and model routing uses γ_i on their respective sets.

Definition 3.1 (Coupled law potential). For fixed $T_q, T_s > 0$, set

$$\begin{aligned}\mathcal{E} = \mathcal{E}_0 &+ \sum_i \left[\sum_{j \in \mathcal{N}_i^q} \beta_{ij} D_{ij}^q + T_q D_{\mathrm{KL}}(\beta_i \| \pi_i^q) \right] \\ &+ \sum_i \left[\sum_{j \in \mathcal{N}_i^s} \gamma_{ij} D_{ij}^s + T_s D_{\mathrm{KL}}(\gamma_i \| \pi_i^s) \right].\end{aligned} \tag{3.4}$$

The reference laws, evaluators, source priors, and temperatures are held fixed in the variations below. The admissible domain consists of laws for which all displayed divergences are finite.

This potential is a constitutive choice at the level of agent laws. No distribution over these law configurations is required to define it. General probability laws enter eq. (3.4); differentiability will require stronger hypotheses. An identity self-channel has $D_{ii}^q = 0$ throughout its domain, with the same convention in the model sector. Independent opposite attending edges remain independent data.

3.2 Source selection and reduction

Proposition 3.2 (Gibbs rows and the exact reduced potential). Hold the agent laws fixed and suppose every source cost is finite. Define

$$\begin{aligned} Z_i^q &= \sum_j \pi_{ij}^q e^{-D_{ij}^q/T_q}, \quad \beta_{ij}^* = \frac{\pi_{ij}^q e^{-D_{ij}^q/T_q}}{Z_i^q}, \\ Z_i^s &= \sum_j \pi_{ij}^s e^{-D_{ij}^s/T_s}, \quad \gamma_{ij}^* = \frac{\pi_{ij}^s e^{-D_{ij}^s/T_s}}{Z_i^s}. \end{aligned} \tag{3.5}$$

These are the unique minimizing rows. With

$$\mathcal{E}_{\mathrm{red}} = \mathcal{E}_0 - T_q \sum_i \log Z_i^q - T_s \sum_i \log Z_i^s, \tag{3.6}$$

the exact identity is

$$\mathcal{E} = \mathcal{E}_{\mathrm{red}} + T_q \sum_i D_{\mathrm{KL}}(\beta_i \| \beta_i^*) + T_s \sum_i D_{\mathrm{KL}}(\gamma_i \| \gamma_i^*). \tag{3.7}$$

Proof. For one row, $\log \beta_j^* = \log \pi_j - D_j/T - \log Z$. Substitute this expression into $T D_{\mathrm{KL}}(\beta \| \beta^*)$ to obtain the row objective plus $T \log Z$. Summing proves eq. (3.7). Relative entropy is nonnegative and vanishes exactly when its two probability laws agree, which proves uniqueness, including competitors on the simplex boundary. □

On a differentiable family, direct differentiation of the finite sums in eq. (3.5) gives

$$d\mathcal{E}_{\mathrm{red}} = d\mathcal{E}_0 + \sum_{ij} \beta_{ij}^* dD_{ij}^q + \sum_{ij} \gamma_{ij}^* dD_{ij}^s. \tag{3.8}$$

Indeed, $-T_q d\log Z_i^q = \sum_j \beta_{ij}^* dD_{ij}^q$, and the other sector is identical. There is no derivative of an optimized row to add. Every dependence of its cost on the laws still belongs in dD. If a source prior were varied, this formula would also contain $-T_q \sum_{ij} \beta_{ij}^* d\log \pi_{ij}^q$ and its model counterpart.

3.3 The derivative includes every interaction role

To display the variation, choose densities against fixed references on a regular dominated family. Write a for the density of a receiver law and b for that of its compared law. For mass-preserving variations and justified differentiation under the integrals,

$$dD_{\mathrm{KL}}(a\|b) = \int \log(a/b)da - \int (a/b)db. \tag{3.9}$$

Here the notation $\int f da$ denotes integration against the signed variation of the law, rather than differentiation of the sample coordinate. The formula follows by differentiating $a\log(a/b)$; the additional integral of da is zero.

For a belief interaction the compared-law variation is

$$d\nu_{ij}^q = (dq_j)\mathcal{K}_{ij}^q[\mathbf{q}] + q_j D\mathcal{K}_{ij}^q[\mathbf{q}](d\mathbf{q}). \tag{3.10}$$

The second term differentiates the prescribed channel itself. Thus an agent contributes when it is a receiver, when it is a source, and whenever its law changes a channel elsewhere. For fixed discrete kernels, the incoming contribution from $j \to i$ to a variation at source value v is

$$-\beta_{ij} \sum_u \frac{q_i(u)}{\nu_{ij}^q(u)} \mathcal{K}_{ij}^q(v,u). \tag{3.11}$$

The model sector has the same expression with q, β replaced by s, γ. If the channels are constructed from law-dependent moments or generators, eq. (3.10) also differentiates those moments and generators. Keeping their numerical values fixed computes a different vector field.

An identity self-channel illustrates why the variable identification matters. Here $a = b$ and $da = db$, so eq. (3.9) is zero throughout the state space. Freezing the source as an old snapshot while moving the receiver replaces this zero functional by a divergence from that snapshot. Its first derivative still vanishes at equality, but its curvature and its force after displacement differ. More generally, directed source selection does not make the gradient of eq. (3.4) one-way: a source appears in the energy of every row that attends to it.

3.4 Fisher dynamics with equal mobilities

Theorem 3.3 (Local flow and exact dissipation on positive tables). Suppose the state and model alphabets are finite, all r_i and evaluator entries $G_i(m,k)$ are strictly positive, and the channel entries are twice continuously differentiable functions of the interior probability tables. Assume every compared-law entry is positive there. Let u^a range over all tables $q_i, s_i, \beta_i, \gamma_i$. Equip their

product with the Fisher metric

$$g_u(v,w) = \sum_a \sum_z \frac{v_z^a w_z^a}{u_z^a}, \qquad \sum_z v_z^a = \sum_z w_z^a = 0. \tag{3.12}$$

For every interior initial condition there is a unique maximal interior solution of

$$\dot{u}_z^a = -u_z^a \left(h_z^a - \sum_w u_w^a h_w^a \right), \qquad h_z^a = \frac{\partial \mathcal{E}}{\partial u_z^a}. \tag{3.13}$$

Along this solution,

$$\frac{d\mathcal{E}}{dt} = -\sum_a \mathrm{Var}_{u^a}(h^a) \le 0. \tag{3.14}$$

Proof. The stated positivity and regularity make $\mathcal{E}$ twice continuously differentiable in interior simplex coordinates. The right side of eq. (3.13) is therefore locally Lipschitz, giving a unique local solution and its maximal continuation in the interior. Its entries sum to zero within each table. Centering h^a removes any ambiguity from extending $\mathcal{E}$ off the normalized tables. For every tangent vector v, the metric pairing of $u^a(h^a - \mathbb{E}_{u^a} h^a)$ with v^a equals $\sum_z h_z^a v_z^a$. This identifies the Fisher gradient. Finally, substitute eq. (3.13) into $\dot{\mathcal{E}} = \sum_{a,z} h_z^a \dot{u}_z^a$ to obtain the sum of negative variances. Interior continuation does not assert that the trajectory stays away from the boundary for all time. □

All blocks have unit mobility. A persistent slow model sector would have to follow from the interactions and their state, rather than from an inserted small multiplier in its equation. One may instead optimize the source rows at each state and apply the same construction to $\mathcal{E}_{\mathrm{red}}$ and the matter tables alone; eq. (3.8) gives its complete derivative.

For a regular finite-dimensional family with positive-definite Fisher tensor $g_{ab}(\theta)$, the corresponding equation is $\dot{\theta}^a = -(g^{-1})^{ab}\partial_b \mathcal{E}$ [1]. Whenever the derivatives exist, contraction gives $\dot{\mathcal{E}} = -(\partial\mathcal{E})^T g^{-1}(\partial\mathcal{E})$. Bimeasurable re-expression of samples, carrying all kernels and reference laws with it, preserves relative entropy and Fisher score inner products. On a family preserved by that re-expression, the gradient therefore transforms as a tangent vector. These statements concern fixed coordinate changes; coordinates that themselves evolve require their coordinate-velocity term.

3.5 What a fixed record model changes

Choose a proper prior $P_0(dy, dJ)$ on represented contents and source labels, and a normalized record kernel $\mathscr{R}(do \mid y, J)$. Then $\mathbb{P} = P_0\mathscr{R}$ is normalized by integrating the record first. At a declared record density $\ell_o(y, J)$ with $0 < Z_o = \int \ell_o dP_0 < \infty$, its posterior is $\Pi_o = \ell_o P_0 / Z_o$. Substitution proves

$$\mathcal{F}(Q; o) = D_{\mathrm{KL}}(Q \| P_0) - \mathbb{E}_Q \log \ell_o = D_{\mathrm{KL}}(Q \| \Pi_o) - \log Z_o, \tag{3.15}$$

when the split terms are defined. The right side supplies the extended definition otherwise. This is the variational free energy of a fixed model. A product family for Q is permitted and its block updates remain coupled through the likelihood factors. Each block derivative contains every factor incident on it.

With independent source priors and one likelihood factor reading each label, a factorized source row has cost $C_{ij} = \mathbb{E}_{Q_Y}[-\log \ell_i(o_i \mid Y, j)]$ and optimizer $\beta_{ij} \propto \pi_{ij} e^{-C_{ij}}$. This expected log likelihood differs from both the peer divergence and the likelihood integrated over a prior. For an isolated source record, exact source inference gives $\mathbb{P}(J = j \mid o) \propto \pi_j \mathbb{E}_{P_Y} \ell_j(o \mid Y)$. Equality of these different rows needs additional hypotheses.

Proposition 3.4 (Entropy and the likelihood Jensen gap). On standard Borel spaces, let $k(w \mid v)$ be a density of a normalized kernel into a receiver space with sigma-finite reference λ. Let the receiver law be $q = a\lambda$ and the source law be p. Put $b(w) = \int k(w \mid v)p(dv)$. Assume $0 < b(w) < \infty$ for q-almost every w, $k > 0$ almost everywhere under $q \otimes p$, and finiteness of $\mathbb{E}_q|\log a|$, $\mathbb{E}_q|\log b|$, and $\mathbb{E}_{q\otimes p}|\log k|$. Define $p^w(dv) = k(w \mid v)p(dv)/b(w)$ on this full-measure set. Then

$$\mathbb{E}_{q\otimes p}[-\log k(W \mid V)] = H_\lambda(q) + D_{\mathrm{KL}}(q\|pK) + \int D_{\mathrm{KL}}(p\|p^w)q(dw). \tag{3.16}$$

Proof. The density ratio gives $D_{\mathrm{KL}}(p\|p^w) = \log b(w) - \int \log k(w \mid v)p(dv)$. Integrating against q and using $-\int \log b dq = H_\lambda(q) + D_{\mathrm{KL}}(q\|pK)$ proves the identity. The last term vanishes exactly when $p^w = p$ for q-almost every w. □

The entropy here is relative to the declared reference. A record likelihood normalized in o need not be normalized in the receiver coordinate w, so proposition 3.4 requires that additional kernel interpretation. Even when it applies, the entropy and conditional divergence can affect the dynamics.

There is also an exact obstruction to universal equality. Freeze a channel K and receiver q. Suppose p_1K, p_2K have positive densities that differ on a set of positive q-measure, with both divergences finite. Strict convexity of $-\log$ gives

$$D_{\mathrm{KL}}(q\|(tp_1 + (1-t)p_2)K) < tD_{\mathrm{KL}}(q\|p_1K) + (1-t)D_{\mathrm{KL}}(q\|p_2K), \qquad 0 < t < 1. \tag{3.17}$$

This follows pointwise on the differing-density set and then by integration. By contrast, the expectation $\mathbb{E}_{q\otimes p}e(W, V)$ of any fixed integrable sample energy is affine in p. It cannot equal the divergence at both endpoints and their strict mixture. The obstruction concerns a fixed sample energy; it does not exclude a direct law potential, a frozen-template episode, or an explicitly different probability model whose variables are law points.

Chapter 4

Prediction and Selective Communication

A model law is useful through the records it predicts. Communication can preserve those predictions while discarding other distinctions, and a receiver can combine a message with information already present in its model. This chapter gives exact identities for both operations. The retained prediction is always named; agreement about one output is not automatically agreement about every future interaction.

4.1 Model presentations and their evaluators

Let $G_i : \mathsf{M}_i \rightsquigarrow \mathsf{K}_i$ be a normalized local model and let $H_i : \mathsf{K}_i \rightsquigarrow \mathsf{O}_i$ be a normalized record kernel. Their composition

$$K_i^{\mathrm{rec}}(m, A) = \int H_i(k, A) G_i(m, dk), \qquad p_i^{\mathrm{rec}} = s_i K_i^{\mathrm{rec}}, \tag{4.1}$$

gives the model's predictive record law. An evaluator may also take a preparation, accessible history, or chosen interaction as input; these are held fixed when two predictions are compared. More general normalized evaluators need not factor through G_i and H_i.

Suppose a model link $\Theta_{ij}^s : \mathsf{M}_j \to \mathsf{M}_i$ is a bimeasurable bijection. Exact compatibility of the two evaluators, after identifying their record alphabets, means

$$K_i^{\mathrm{rec}}(\Theta_{ij}^s m, A) = K_j^{\mathrm{rec}}(m, A). \tag{4.2}$$

Consequently, $s_i = (\Theta_{ij}^s)_{\#} s_j$ implies equal predictive record laws. The compatibility is additional information: two equal laws on a one-point model space can predict opposite deterministic bits through different evaluators.

Theorem 4.1 (Exact predictive information identity). Let s, t be probability laws on a standard Borel model space and let K be a common normalized record kernel. Assume $D_{\mathrm{KL}}(s\|t) < \infty$. Set $\mu(dm, do) = s(dm)K(m, do)$ and $\nu(dm, do) = t(dm)K(m, do)$, and denote their conditional model laws given o by s^o, t^o. Then

$$D_{\mathrm{KL}}(s\|t) = D_{\mathrm{KL}}(sK\|tK) + \int D_{\mathrm{KL}}(s^o\|t^o)(sK)(do). \tag{4.3}$$

Proof. The common kernel cancels in the joint Radon–Nikodym derivative, giving $D_{\mathrm{KL}}(\mu\|\nu) = D_{\mathrm{KL}}(s\|t)$. Disintegrating the same joint laws by their record marginal factors their density ratio into

the record ratio and the conditional model ratio. The logarithm is their sum. Integrating the first term gives $D_{\mathrm{KL}}(sK\|tK)$, and integrating the second first over m gives the displayed conditional term. Finiteness makes the decomposition unambiguous. □

The second term is the model distinction that the record leaves unresolved. Its vanishing is an exact sufficiency condition. A constant evaluator can make all predictions agree while retaining no model distinction at all. Useful agreement therefore requires a specified predictive task as well as a comparison of laws. For a common history kernel, the same identity holds with the full finite record history in place of o.

If the model link depends on the current laws, theorem 4.1 applies to the resulting fixed comparison at that configuration, provided eq. (4.2) holds there. Preservation of compatibility during an update is a separate condition on the dynamics and the evaluator.

4.2 What the receiver reconstructs

A sending map $T_j : E_j \to \mathsf{H}$ induces the message law $\nu_j = (T_j)_{\#}q_j$. Different laws can have the same message law. For linear maps $T_j : E_j \to \mathsf{H}$ and $O_i : \mathsf{H} \to E_i$,

$$\operatorname{rank}(O_i T_j) \le \operatorname{rank}(T_j). \tag{4.4}$$

Writing a message into a larger space cannot by itself recover distinctions discarded by T_j. A richer reconstruction uses the receiver's conditional model.

For a received value h and a declared likelihood $\ell_i(h \mid k)$, Bayesian assimilation is

$$q_i^+(dk) = \frac{\ell_i(h \mid k)\, q_i^-(dk)}{\int \ell_i(h \mid k')\, q_i^-(dk')}, \tag{4.5}$$

when the denominator is finite and positive. An unmeasured coordinate can change because it is conditionally related to the measured one in q_i^-. The observation selects among possibilities already related by the receiver's model.

Proposition 4.2 (Exact conditional reconstruction). Let $T : E \to \mathsf{H}$ be measurable on standard Borel spaces. Let q, p be laws with $D_{\mathrm{KL}}(q\|p) < \infty$, and set $\nu = T_{\#}q$. Choose a regular conditional law $p(dk \mid h)$ for p given $T = h$. Define

$$\widehat{q}(dk) = \int p(dk \mid h)\, \nu(dh). \tag{4.6}$$

Then $T_{\#}\widehat{q} = \nu$ and

$$D_{\mathrm{KL}}(q\|\widehat{q}) = \int D_{\mathrm{KL}}\big(q(\cdot \mid h)\|p(\cdot \mid h)\big)\, \nu(dh). \tag{4.7}$$

In particular, $\widehat{q} = q$ exactly when the two conditional laws agree for ν-almost every h.

Proof. Absolute continuity gives $\nu \ll T_{\#}p$, so the conditional version of p is defined at every h needed up to a ν-null set. That conditional is supported on $T^{-1}(h)$ almost surely. Hence its mixture has pushforward ν. The laws q and $\widehat{q}$ have the same h-marginal and respective conditional laws

$q(\cdot \mid h)$ and $p(\cdot \mid h)$. Factoring their density ratio by h proves eq. (4.7). Nonnegativity and the equality condition of relative entropy give the last claim. □

This reconstruction preserves the communicated marginal and supplies the discarded conditional from p. It makes precise the sense in which a receiver contributes structure that the message alone does not specify. The proposition assumes a common content presentation; a link or a separately declared correspondence is needed before applying it across different fibers.

4.3 A predictive coding criterion

Lossy source coding makes fidelity part of the problem definition [5]. Decoder side information is a realized variable correlated with the source, as in Wyner–Ziv coding [9]; knowing a prior without such a realization does not supply that variable. A few noiseless real numbers also do not specify a finite-bit channel. A communication rate requires a declared coding, precision, or noise model.

For a predictive criterion, let $P(X, Y, Z)$ be a declared preparation law. The source X may be represented content or a complete fine configuration; the choice is fixed in each application. Let Z be accessible receiver information and let Y be a response to be preserved. An encoder $T(du \mid X)$ gives the conditional independence

$$U - X - (Y, Z). \tag{4.8}$$

The decoder predicts Y from (U, Z). This is the setting of an information bottleneck, with the relevant variable explicitly named [6].

Theorem 4.3 (Predictive distortion and the optimal decoder). On standard Borel spaces, assume the displayed information quantities are finite and eq. (4.8) holds. For any predictive decoder $D(dy \mid u, z)$ with finite expected relative entropy,

$$\begin{aligned} &\mathbb{E} D_{\mathrm{KL}}\big(P(Y \mid X, Z) \| D(Y \mid U, Z)\big) \\ &\quad = I(Y; X \mid U, Z) + \mathbb{E} D_{\mathrm{KL}}\big(P(Y \mid U, Z) \| D(Y \mid U, Z)\big). \end{aligned} \tag{4.9}$$

The optimal decoder is $P(dy \mid u, z)$. Its distortion is

$$\mathcal{D}(T) = I(Y; X \mid U, Z) = I(Y; X \mid Z) - I(Y; U \mid Z). \tag{4.10}$$

Proof. Conditional independence gives $P(Y \mid X, U, Z) = P(Y \mid X, Z)$. In the logarithmic ratio defining the left side of eq. (4.9), insert the intermediate conditional law $P(Y \mid U, Z)$. The first resulting expectation is the definition of $I(Y; X \mid U, Z)$. In the second, averaging over X gives the conditional law $P(Y \mid U, Z)$, yielding the remaining divergence. The second term vanishes at the stated decoder and is nonnegative for every decoder. Finally, expand $I(Y; X, U \mid Z)$ in both orders and use $I(Y; U \mid X, Z) = 0$. □

The functional $I(X; U \mid Z) + \lambda \mathcal{D}(T)$, for $\lambda \geq 0$, is one possible criterion for selecting a representation. Its first term is an information cost, not automatically the length of an individual message. An operational coding theorem requires its own source and block-coding assumptions. Transmitting a sample from $(T_j)_{\#} q_j$ and transmitting that whole probability law are also different protocols.

For parent formation, choose X to include the fine configuration and choose Y to be a future boundary response over a specified horizon and interaction. Zero distortion means that U and Z are sufficient for this target under this preparation. A subsequent interaction may read information that this target ignored. Exact inheritance therefore requires the dynamical condition developed in chapter 5, not only predictive sufficiency for one chosen response.

Chapter 5

Inherited Parents and Recursive Formation

A collection becomes a parent when its retained state supports inherited interaction and evolution. A probability law obtained by compression is a first part of this construction. It does not determine how that law will change, what the parent can communicate, or which information its children must retain. We give exact conditions for these further steps and a finite example in which inheritance works through two levels. The example supplies its reductions; formation of the reductions remains a separate question.

5.1 Two objects that a parent must retain

Definition 5.1 (Sample compression and configuration retention). Let $x \in \mathsf{S}$ be a complete agent configuration and let Q_x be its declared law on a standard Borel space Y of represented contents. A normalized sample channel $C : \mathsf{Y} \rightsquigarrow \mathsf{Z}$ gives the retained law

$$Q_x^c(A) = \int C(y, A) Q_x(dy). \tag{5.1}$$

A configuration retention is a measurable map $R : \mathsf{S} \to \mathsf{S}^c$. Its value includes the parent laws and the additional data needed for its declared interactions and update.

The distinction applies to both sectors. A parent carrying q_I and s_I also needs the evaluator that makes s_I predictive. Boundary channels, source rows, source marks, and memory can be further components of $R(x)$. Equal marginal laws do not imply equality of these components. When a sample channel reads random links or source variables, those variables belong to the joint input law of the channel. Substitution of a mean link or a selected source is a different retention.

A law of represented contents and a probability law over complete configurations are different objects. Coupled deterministic evolution of law-valued agents requires no probability distribution over x. If such a distribution is used below, its preparation or transition mechanism is stated. All constructions admit general probability laws. A Gaussian parent is one possible family, subject to its own closure conditions.

5.2 Exact deterministic inheritance

Proposition 5.2 (An update descends precisely when it respects retained states). Let S, S^c be measurable spaces, let $R : \mathsf{S} \to \mathsf{S}^c$ be a measurable surjection with a measurable right inverse ι,

and let $F : \mathsf{S} \to \mathsf{S}$ be measurable. There is a measurable parent update F^c satisfying

$$R \circ F = F^c \circ R, \tag{5.2}$$

if and only if

$$R(x) = R(x') \quad \Longrightarrow \quad R(F(x)) = R(F(x')). \tag{5.3}$$

The parent update is unique. Finite spaces automatically admit the required right inverse.

Proof. Equation (5.2) implies the implication in (5.3). Conversely, define $F^c = R \circ F \circ \iota$. It is measurable. Since $R(\iota(R(x))) = R(x)$, the implication gives $F^c(R(x)) = R(F(x))$. Every parent state has a preimage, so this equation also fixes F^c uniquely. □

Repeated substitution gives $R \circ F^n = (F^c)^n \circ R$ for every nonnegative integer n. This is an exact statement about the chosen update, including its order and retained parameters. Replacing that update by descent of a newly defined parent objective requires another argument.

For a differentiable version, suppose R is continuously differentiable and the fine and parent vector fields B, B^c have unique local solutions. Their flows satisfy $R \circ \Phi_t = \Phi^c_t \circ R$ on their common existence interval if

$$DR_x B(x) = B^c(R(x)). \tag{5.4}$$

Indeed, the chain rule makes $R(\Phi_t(x))$ a parent solution with initial value $R(x)$, and uniqueness identifies it. Conversely, differentiation of the flow identity at zero gives (5.4). For a moving map R_t, the left side becomes $\partial_t R_t(x) + DR_t(x)B_t(x)$. Membership changes therefore need an explicit handoff map; a differential identity on one fixed partition does not specify them.

Definition 5.3 (Loss of dynamical roles under product recognition). Take $Q_x = \mathrm{Ber}(p_1) \otimes \mathrm{Ber}(p_2)$, with $x = (p_1, p_2) \in [0,1]^2$, and the coupled deterministic law

$$\dot{p}_1 = -p_1 + \lambda(p_2 - p_1), \qquad \dot{p}_2 = -2p_2 + \lambda(p_1 - p_2), \qquad \lambda \geq 0. \tag{5.5}$$

The linear system has a unique global solution and its vector field points inward on every face of the square. Retain only the law of $K_1 + K_2$. The configurations $(0.8, 0.2)$ and $(0.2, 0.8)$ both give probabilities $(0.16, 0.68, 0.16)$ for the values $0, 1, 2$. However,

$$\frac{d}{dt}\mathbb{E}(K_1 + K_2) = -(p_1 + 2p_2), \tag{5.6}$$

equals -1.2 and -1.8 respectively. Their retained laws therefore have different derivatives. No autonomous update of that retained law can reproduce both trajectories.

This obstruction occurs while recognition remains a product. The missing datum is which probability belongs to which dynamical role. Correlated recognition is neither a prerequisite for this counterexample nor its automatic remedy. Coupled mean-field dynamics are admissible; the proposed retention must still pass its own inheritance test.

5.3 Exact stochastic inheritance

Proposition 5.4 (Finite strong lumpability). Let T be a transition matrix on a finite configuration space S and let $R : \mathsf{S} \to \mathsf{S}^c$ be surjective. A single parent transition matrix reproduces the retained process for every initial law if and only if, for every $z' \in \mathsf{S}^c$,

$$\sum_{y:R(y)=z'} T(x,y) = \sum_{y:R(y)=z'} T(x',y) \quad \text{whenever } R(x) = R(x'). \tag{5.7}$$

In that case the common sum defines $T^c(R(x), z')$ uniquely.

Proof. The common sums are nonnegative and sum to one, so they define a transition matrix. Conditional on any retained history ending at z, the hidden configuration lies in $R^{-1}(z)$. Every state in that fiber gives the same next retained distribution $T^c(z,\cdot)$; averaging over its conditional law leaves this distribution unchanged. Induction gives the parent path law from the pushed initial law. Conversely, start the fine process at each point mass δ_x. A common parent kernel must give the same one-step distribution whenever these initial states have the same image. This is (5.7). □

This is the finite strong-lumpability criterion [3]. On standard Borel spaces the same argument applies when $T(x, R^{-1}A)$ factors through R as a measurable probability kernel. The qualification is part of the construction. Closure for every initial law is stronger than closure for one chosen preparation.

An equilibrium conditional average of fine transitions defines a normalized candidate parent kernel. It does not prove (5.7): the hidden conditional law during a transient can differ from the equilibrium conditional. Likewise, agreement of invariant distributions does not establish agreement of histories.

5.4 Boundary responses and composition

A parent must inherit what it does to the rest of the population. Fix a declared boundary protocol a, including the preparation and receiving conditions. Let $K_a : \mathsf{S} \rightsquigarrow \mathsf{O}_a$ be its normalized response kernel and let $D_a : \mathsf{O}_a \rightsquigarrow \mathsf{O}_a^c$ retain the response of interest. An inherited kernel K_a^c satisfies

$$(K_a D_a)(x, A) = K_a^c(R(x), A). \tag{5.8}$$

Here $(KD)(x,A) = \int D(o,A)K(x,do)$. Under the measurable right-inverse hypothesis of proposition 5.2, such a kernel exists exactly when $K_a D_a$ is constant on each fiber of R. Define it using the right inverse to prove sufficiency; substitution proves necessity. When several outputs are used jointly, this condition applies to their joint kernel. Preserving each output marginal alone need not preserve their dependence.

For a child channel $\mathcal{K}_{ij} : \mathsf{Y}_j \rightsquigarrow \mathsf{Y}_i$ and retentions $C_j : \mathsf{Y}_j \rightsquigarrow \mathsf{Z}_J$ and $C_i : \mathsf{Y}_i \rightsquigarrow \mathsf{Z}_I$, the corresponding square is $\mathcal{K}_{ij} C_i = C_j \mathcal{K}_{IJ}^c$, where $\mathcal{K}_{IJ}^c : \mathsf{Z}_J \rightsquigarrow \mathsf{Z}_I$. The rightmost expression means first retain the sender and then apply the parent channel. The retained sender must determine every distinction visible to the retained receiver. In a linear realization this reduces to $C_i M_{ij} = M_{IJ}^c C_j$; the kernel-inclusion criterion and its proof are given in the supplement.

Proposition 5.5 (Inheritance composes). Suppose $R_{10} : \mathsf{S}_0 \to \mathsf{S}_1$ and $R_{21} : \mathsf{S}_1 \to \mathsf{S}_2$ intertwine deterministic updates at successive levels. Then $R_{20} = R_{21} \circ R_{10}$ intertwines the first and third updates. The analogous assertion holds for finite lumped Markov transitions. Let $K_a^\ell : \mathsf{S}_\ell \rightsquigarrow \mathsf{O}_\ell$ be boundary kernels, with response retentions $D_{10} : \mathsf{O}_0 \rightsquigarrow \mathsf{O}_1$ and $D_{21} : \mathsf{O}_1 \rightsquigarrow \mathsf{O}_2$. If

$$K_a^0 D_{10} = R_{10} K_a^1, \qquad K_a^1 D_{21} = R_{21} K_a^2, \tag{5.9}$$

where maps on the right are their deterministic kernels, then $K_a^0 D_{10} D_{21} = R_{10} R_{21} K_a^2$.

Proof. For deterministic updates, $R_{21} R_{10} F^0 = R_{21} F^1 R_{10} = F^2 R_{21} R_{10}$, where juxtaposition in this identity denotes function composition. For Markov transitions and their deterministic retention kernels, $T^0 R_{10} R_{21} = R_{10} T^1 R_{21} = R_{10} R_{21} T^2$. For boundary kernels, associate the successive integrals and substitute (5.9). All integrands for events are nonnegative, so interchanging the integration order is valid. □

When changes of gauge presentation act on the retained spaces, the retentions and boundary kernels must transform compatibly with those actions. Re-expressing a commuting square then gives the same square in the new coordinates. Neither a parent frame chosen for this purpose nor the square itself determines a base connection.

5.5 A finite two-level construction

Let a complete configuration be $x = (a, b, c, r) \in \{0, 1\}^4$. The coordinates are labels of internal configurations, with r an internal binary register and $\rho(x) = r$ its readout. To attach actual agent laws, choose

$$q_a = \mathrm{Ber}\left(\frac{1 + 2a}{4}\right), \qquad s_a = \mathrm{Ber}\left(\frac{2 + a}{4}\right), \qquad \zeta_a = q_a \otimes s_a. \tag{5.10}$$

Thus both sectors vary, with tied parameters in this supplied family; b, c are additional internal marks. A model presentation $m \in \{0, 1\}$ has evaluator

$$K(m, \{1\}) = \frac{1}{4} + \frac{m}{2}. \tag{5.11}$$

The corresponding predictive law is $\nu_a = s_a K = \mathrm{Ber}(1/2 + a/8)$, which distinguishes the two values of a.

At each step draw independent bits $\xi \sim \mathrm{Ber}(1/4)$, $\eta \sim \mathrm{Ber}(1/3)$, and $\upsilon \sim \mathrm{Ber}(1/5)$, independently of preceding draws, and set

$$a' = a \oplus \xi, \qquad b' = b \oplus a \oplus \eta, \qquad c' = c \oplus b \oplus \upsilon. \tag{5.12}$$

Independently of these bit draws, conditional on the current configuration, give r' law ν_a. This supplies a transition of an internal register; the evaluator alone did not force this choice. The symbol $\oplus$ denotes addition modulo two. The dynamics are coupled: a affects the next b, and b affects the next c.

Retain $R_{10}(a,b,c,r) = (a,b,r)$ and then $R_{21}(a,b,r) = (a,r)$. For $f_p(0) = 1-p$ and $f_p(1) = p$, the first parent kernel is

$$T^1((a,b,r),(a',b',r')) = f_{1/4}(a' \oplus a)\nu_a(\{r'\}) \cdot f_{1/3}(b' \oplus b \oplus a). \tag{5.13}$$

Summing the fine transition over c' gives this expression, independently of c. Summing it over b' gives $T^2((a,r),(a',r')) = f_{1/4}(a' \oplus a)\nu_a(\{r'\})$, independently of b. Both reductions satisfy proposition 5.4. Their configuration spaces have sizes sixteen, eight, and four, and both parents retain the laws in (5.10), their evaluator, and the register. Its readout descends by the identity on record values. The first parent additionally retains b.

Both parents reproduce the entire finite record history as well as the retained configuration history. Indeed, write the fine path probability as its initial mass times its transition factors. Summing over hidden coordinates gives the parent path law by proposition 5.4; the record at each step is already a coordinate of that path. The next record has probability $1/2 + a/8$ of being one, so the inherited response is nonconstant. No independence of the initial coordinates is needed.

This construction has nontrivial reductions and coupled dynamics, but its retentions were specified in advance. It proves exact two-level inheritance, without proving that the dynamics choose either reduction.

5.6 Memory and formation

When a proposed retention fails, one exact option is to retain more of the configuration. For a declared stochastic process, another is to retain the conditional law of the hidden configuration given the accessible retained history. That conditional law determines future predictions. In a finite model it is a probability vector with an exact recursive update, derived in the supplement. Its attainable values need not form a finite set. Introducing such a filter presupposes a process and a preparation law; it is not compulsory for deterministic coupled agents.

A handoff initializes the parent from the actual fine state or its declared conditional law, including the source marks and memory read by future interactions. When evidence is shared across levels, its likelihood belongs to the joint model once. Reusing a child's record through a complete same-episode parent posterior requires the appropriate conditional or cavity accounting. A subsequent predictive prior is a different operation.

Open problem 5.6 (Inherited recursive formation). Construct a specified population dynamics that selects nontrivial blocks from an admitted class of initially unpartitioned configurations, with a declared persistence criterion and a handoff determined by the state or accessible history. Each selected block must carry inherited belief and model laws, evaluators, boundary responses, and dynamics. The resulting parents must then select a block containing at least two first-stage parents under those inherited interactions, without resetting their source data or supplying a newly tuned attraction at the next level.

Persistence can mean an invariant block relation in deterministic dynamics or a specified path event in stochastic dynamics. The finite construction settles inheritance for a supplied hierarchy. It does not select that hierarchy or establish persistence.

Chapter 6

The Remaining Formation Problem

The constructions establish a language for agents that compare laws, receive internal records, and inherit descriptions across changes of scale. They also identify what a proposed hierarchy must preserve. A normalized predictor makes model agreement operational; an inherited update makes a retained description dynamical. These requirements are logically separate from identifying a coherent block at one instant.

The quantifiers in open problem 5.6 matter. A prescribed partition can admit exact dynamics without being selected by the dynamics. A block selected at one time can fail to persist. A parent can reproduce one record law and fail to reproduce the next response after an interaction. A smaller model-sector mobility can impose a timescale separation without explaining its origin. Each is a distinct intermediate construction, and each can be useful without resolving the formation problem.

The next exact example should retain both q and s, give the model presentations a normalized evaluator, and specify the law-dependent interfaces. Candidate parent data should then be tested for sufficiency by comparing fine states with identical retained data. If their future retained states differ, the example itself identifies information that the parent must keep. That information may be a conditional dependence, a dynamical role, an interface response, or memory of a shared source. It need not be a covariance, and it need not fit a predetermined distribution family.

The same requirement constrains the eventual base geometry. An agent can support a curved connection when it has compatible directional transports. A state-dependent interaction between agents at one context does not determine those transports by itself. Deriving them requires a rule relating nearby contexts and its patch compatibility. A common geometry across agents requires a further compatibility statement. Neither task is made automatic by close model laws.

Appendix A

A Normalized Internal Record Process

This appendix constructs a finite-history law from an internal interaction rule. The rule is supplied; normalization does not select it. Let S and O be standard Borel spaces. A state x includes an accessible record register. Write $\rho : \mathsf{S} \to \mathsf{O}$ for its readout.

Definition A.1 (Event and registration). Choose a normalized kernel $R : \mathsf{S} \rightsquigarrow \mathsf{O}$ and a measurable update $A : \mathsf{S} \times \mathsf{O} \to \mathsf{S}$ satisfying

$$\rho(A(x, o)) = o, \tag{A.1}$$

for $R(x, \cdot)$-almost every o, for every admitted x. Define

$$J(x, dx', do) = R(x, do)\delta_{A(x,o)}(dx'). \tag{A.2}$$

If necessary, the state can be explicitly augmented by a register so that eq. (A.1) holds.

Proposition A.2 (Consistent finite histories). For an initial law μ_0 on S, the iterated kernel product

$$\mathbb{P}_n(dx_0, do_1, dx_1, \ldots, do_n, dx_n) = \mu_0(dx_0) \prod_{t=1}^{n} J(x_{t-1}, dx_t, do_t), \tag{A.3}$$

is a probability law. Marginalizing its last pair gives $\mathbb{P}_{n-1}$, and every record agrees with the readout of the resulting state almost surely.

Proof. The product denotes successive kernel integration, not independent factors. Integrating the last pair gives one because $J(x, \mathsf{S} \times \mathsf{O}) = 1$. Repeating this argument gives total mass one and the stated consistency. Each transition is supported on $x' = A(x, o)$ and hence on $\rho(x') = o$. For a finite history, the intersection of these probability-one events still has probability one. □

Nothing in this construction requires a variable external to the agents. It does require a declared accessible register and an update. A stochastic A is handled by replacing its Dirac kernel with a normalized kernel supported on the registration condition. A deterministic transition is recovered when R is a Dirac kernel as well.

A.1 Prediction followed by assimilation

One possible internal state contains a model law s and a record register. Fix a normalized density $k(o \mid m)$ with respect to a reference measure on O. Let

$$R(s, do) = \left[\int k(o \mid m)s(dm)\right] do, \qquad s^o(dm) = \frac{k(o \mid m)s(dm)}{\int k(o \mid m')s(dm')}. \tag{A.4}$$

The update replaces s by s^o and stores o. The denominator is positive and finite for predictive-almost every record. Defining an arbitrary probability law on the remaining null set completes the measurable rule under the usual conditional-law hypotheses.

Proposition A.3 (The Bayesian history identity). For a fixed latent model m, initial law s_0, and fixed normalized observation density k, sequential prediction and Bayesian assimilation give the record density

$$p(o_1, \dots, o_n) = \int \prod_{t=1}^{n} k(o_t \mid m) s_0(dm). \tag{A.5}$$

Proof. After $t-1$ records, the posterior is proportional to $s_0(dm) \prod_{r<t} k(o_r \mid m)$. The next predictive density is the integral with the product extended to t, divided by the integral up to $t-1$. Multiplying these conditional densities cancels every intermediate normalizer and leaves eq. (A.5). □

This identity applies to the stated fixed observation model. If the likelihood itself is changed using the current state, the product of the actual predictive kernels still defines a normalized process, but it need not equal the fixed-latent expression. Likewise, replacing Bayes' rule by a finite natural-gradient step defines a different assimilation mechanism unless the updates coincide. The normalization of a process and its interpretation as Bayesian conditioning are separate statements.

Appendix B

From Directional Interaction to Base Transport

Links between agents at a single context do not determine transport between neighboring contexts. A connection requires a smooth dependence on the direction and size of a base displacement. This construction can occur within one agent; it does not require collective averaging.

Fix one sector and one agent, suppressing the sector superscript. On a coordinate neighborhood of $\mathcal{C}$, let $U_i(c)$ be a twice continuously differentiable invertible presentation and let $a_i = a_{i,\mu}(c)dc^\mu$ be a continuously differentiable matrix-valued one-form. For a principal G-connection its values must lie in the represented Lie algebra. General endomorphism-valued forms instead describe connections for the extended general linear frame bundle.

Definition B.1 (Infinitesimal directional transport). The proposed transport family has first-order behavior

$$\Theta_i(c+\delta c, c) = U_i(c+\delta c)\exp[-a_{i,\mu}(c)\delta c^\mu]U_i(c)^{-1} + O(|\delta c|^2). \tag{B.1}$$

The first-order generator is linear in the tangent displacement. A single finite matrix at c supplies neither this linear dependence nor the identity value at zero displacement.

Proposition B.2 (Connection and curvature). With the convention $D_i = d + B_i$, eq. (B.1) has connection and curvature

$$B_i = U_i a_i U_i^{-1} - dU_i U_i^{-1}, \qquad F_i = U_i(da_i + a_i \wedge a_i)U_i^{-1}. \tag{B.2}$$

Proof. Expanding the two factors at c gives $\Theta_i = I + (dU_iU_i^{-1} - U_ia_iU_i^{-1})(\delta c) + O(|\delta c|^2)$, which identifies B_i. Acting on a vector-valued function gives the operator identity $d + B_i = U_i(d + a_i)U_i^{-1}$. Squaring it yields $(d + B_i)^2 = U_i(d + a_i)^2U_i^{-1}$, and the squares act by the curvature two-forms in eq. (B.2). □

For an explicit curved example, take $U_i = I$ over an open subset of $\mathbb{R}^2$ and set

$$a_{i,x} = E_{12}, \qquad a_{i,y} = E_{21}. \tag{B.3}$$

Both are rank-one receiving–sending products, while their derivatives vanish. Nevertheless,

$$F_{i,xy} = [E_{12}, E_{21}] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \neq 0. \tag{B.4}$$

Thus even constant directional interfaces can give curvature through noncommutativity. By contrast, replacing a finite pointwise map by a new frame $V(c)$ and setting $B = -dVV^{-1}$ gives zero curvature: it is the case $a = 0$ of eq. (B.2).

Finite paths use the solution of

$$\frac{d}{dt}T_i(t) = -B_i(\dot{\gamma}(t))T_i(t), \qquad T_i(0) = I. \tag{B.5}$$

Uniqueness of this linear equation gives composition under concatenation and inversion under path reversal. The solution is the path-ordered exponential; the usual link and Wilson-line constructions use this ordering [7]. Independently declared opposite attending edges are not two orientations of this same path and need not be inverses.

Proposition B.3 (Compatibility under re-expression). Under an agent-coordinate change $U_i' = g_i U_i$ with the same reference form,

$$B_i' = g_i B_i g_i^{-1} - dg_i g_i^{-1}. \tag{B.6}$$

Under a change of reference presentation $U_i' = U_i h$, keeping the same intrinsic transport instead requires

$$a_i' = h^{-1} a_i h + h^{-1} dh. \tag{B.7}$$

Proof. Insert $g_i U_i$ into eq. (B.2) and apply the product rule. Inserting $U_i h$ and eq. (B.7) cancels the two terms containing dh, leaving the original B_i. □

The same affine law must hold between overlapping coordinate patches. A directional interface product transforming only by conjugation is a tensorial one-form. It can describe the difference between two connections, but it is not by itself the full affine connection data. Constructing displacement links with the endpoint transformation law and then differentiating them is one way to obtain the required inhomogeneous term.

If patch coordinates satisfy $v_\alpha = g_{\alpha\beta} v_\beta$, their transition functions must obey $g_{\alpha\beta} g_{\beta\gamma} = g_{\alpha\gamma}$ on triple overlaps. The displayed affine law then makes the local connections compatible. A nontrivial bundle need not admit a single global presentation U_i. Even on a trivial bundle, zero curvature does not imply path independence: the scalar connection $B = b\,d\theta$ on the circle is flat but has holonomy $e^{-2\pi b}$, which differs from one for real $b \neq 0$. Global path independence requires trivial holonomy as well as local flatness.

Different agents may carry different connections on the same bundle. They represent one common connection only when their local forms satisfy its overlap relations. Neither case requires an averaging construction.

Open problem B.4 (Deriving directional transport from agent interactions). Specify a constitutive family of displacement links from the agents' laws and interfaces that has identity at zero displacement, a smooth first derivative linear in displacement, and compatible endpoint laws on overlaps. Its derivative then determines the connection by proposition B.2. The unresolved step is the selection and construction of that family; pointwise covariance and the possibility of curvature alone do not determine it.

Appendix C

Notation

Symbol	Meaning
$\mathcal{C}, c_*$	Context base and a selected context.
P, G	Principal bundle and its structure group.
K, M	Represented state contents and model presentations.
$\mathcal{P}(X)$	Probability measures on a measurable space X.
ζ_i, q_i, s_i	Joint agent law and its state and model marginals.
U_i^q, U_i^s	Frame representatives in the two sectors.
$\bar{q}_i, \bar{s}_i$	Laws in common reference coordinates on a trivializing patch.
L_{ij}^q, L_{ij}^s	Law-dependent interaction generators in the vector realization.
$\Theta_{ij}^q, \Theta_{ij}^s$	Invertible source-to-receiver interaction maps.
$\mathcal{K}_{ij}^q, \mathcal{K}_{ij}^s$	Normalized source-to-receiver comparison channels.
T_j^a, O_i^a	Sending and receiving maps in sector $a \in \{q, s\}$.
β_i, γ_i	Belief-sector and model-sector source rows.
G_i, H_i	Local model-to-state kernel and state-to-record kernel.
K_i^{rec}	Normalized model-to-record predictive evaluator.
S, x	Complete agent-configuration space and a configuration.
O$_i$, ρ_i	Record alphabet and internal record readout.
$O_{ij,t}, o_{ij,t}$	Record random variable and one realized value.
Γ_{ij}	Interaction transition on configurations.
$\mathcal{E}$	Declared law-level interaction potential.
$\mathcal{F}$	Variational free energy for a specified normalized probabilistic model.
F, R, F^c	Fine deterministic update, retention map, and inherited update.
a_i, B_i, F_i	Reference connection form, its local representative, and curvature.
X, U, Z, Y	Source, encoded representation, decoder information, and prediction target in the coding construction.

The symbol F for a discrete update and F_i for curvature are used in separate constructions. A frame U_i^a is an invertible map; the coding variable U is a random representation. A Markov kernel returns a probability measure, whereas an ordinary map returns one value. Kernel composition is written in the order in which a law traverses the kernels.

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